

Data Center Requirements

- Data centers run two types of apps
 - **outward** (user) facing (e.g., serving web pages to users)
 - **internal** computations (e.g., MapReduce for web indexing)
- Workloads often **unpredictable**:
 - Multiple services run concurrently within a DC
 - Demand for new services may spike unexpected
 - Spike of demands for new services mean **success**!
 - But this is when success spells trouble (if not prepared)!
- **Failures** of servers are the norm
 - Resort to dynamic re-assignment of servers, jobs/tasks (to deal with failures; data is often replicated across racks, ...)
 - **Traffic matrix** between servers are constantly changing

Data Center Design Goals

- **Agility** – Any service, Any Server
 - Turn servers into a single large pool
 - Let services “breathe”: dynamically expand and contract their footprint as needed
- Benefits ← **three motivators for most data center infrastructure projects!**
 - More productive service developers
 - Lower cost
 - Higher performance and reliability

How to Achieve Agility

- **Workload** Management
 - rapidly **launching** a service program on a server
 - dynamical **cluster scheduling** and **server assignment**
 - prepare **virtual images**
- **Storage** Management
 - each server has access persistent data
 - (often) distributed file systems
- **Network** Management
 - communicating with other servers: **in** or **outside** of the data center
 - high **performance** and **reliability**

Requirements of (Data Center) Networking

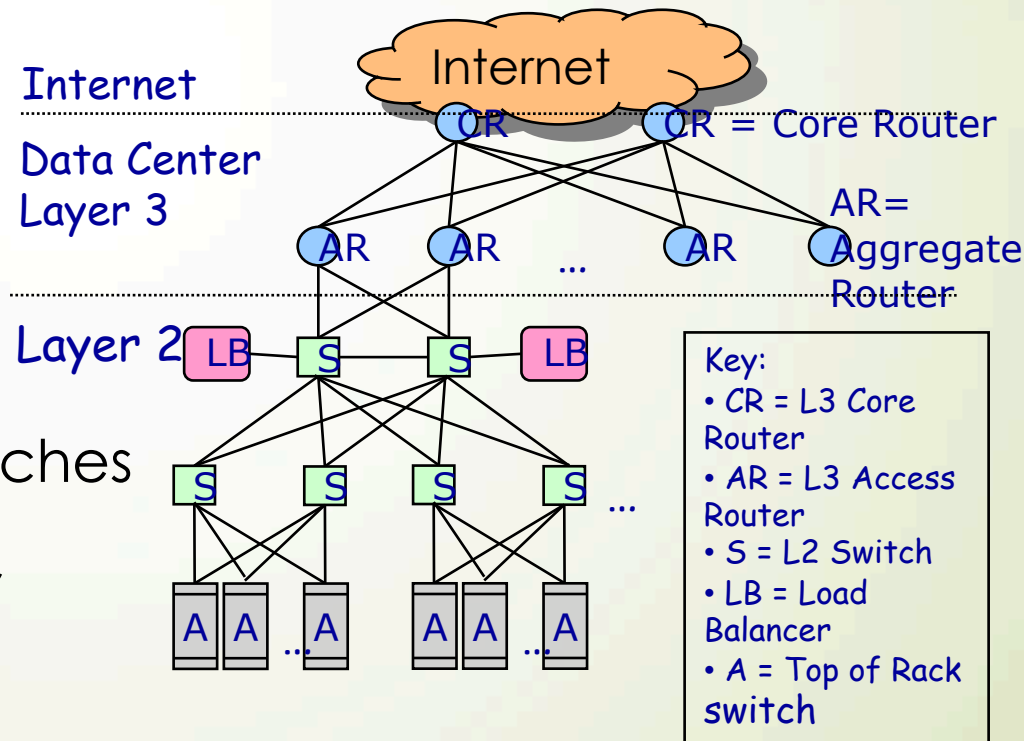
- Uniform high capacity
 - Capacity between (**any two**) servers limited only by their NICs
 - No need to consider topology when adding servers ← **predefined topology**
- Performance isolation
 - Traffic of one service should not be interfered by others ← **multi-tenants**
- Easy to manage: “Plug-&-Play” (layer-2 semantics ← **more later...**)
 - Flat addressing, so any server can have any IP address
 - Server configuration is the same as in a LAN
 - Legacy apps (e.g., depending on broadcast) must work

Agenda

- Modern Data Centers
- Design Goals of Data Center Networks
- (Old) Data Center Networks
- Fat-Tree Network Topology
- Virtual Layer 2 (VL2) Networks

Old Data Center Architecture

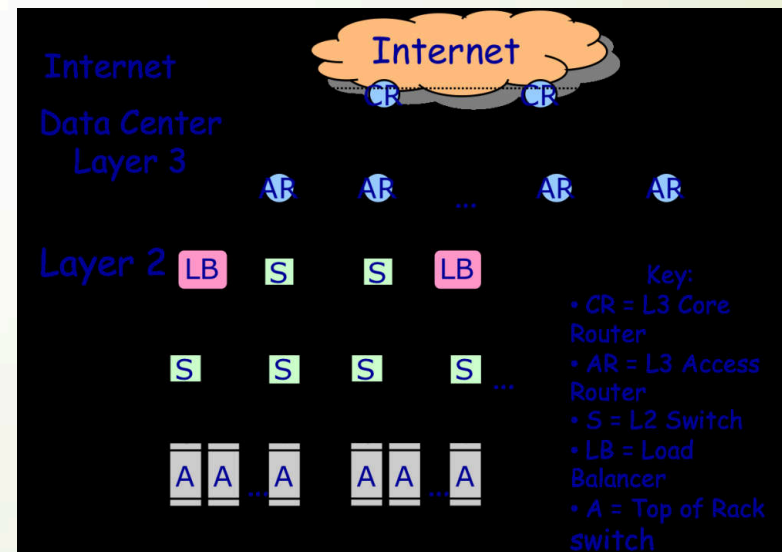
- Hierarchical network
 - 1+1 redundancy
- Switches higher in the hierarchy handles more traffic
 - more expensive, more efforts made at availability → **scale-up design**
- Servers connect via 1 Gbps category-5 cable to Top-of-Rack (ToR) switches
- Other links are mix of 1G, 10G; fiber, copper



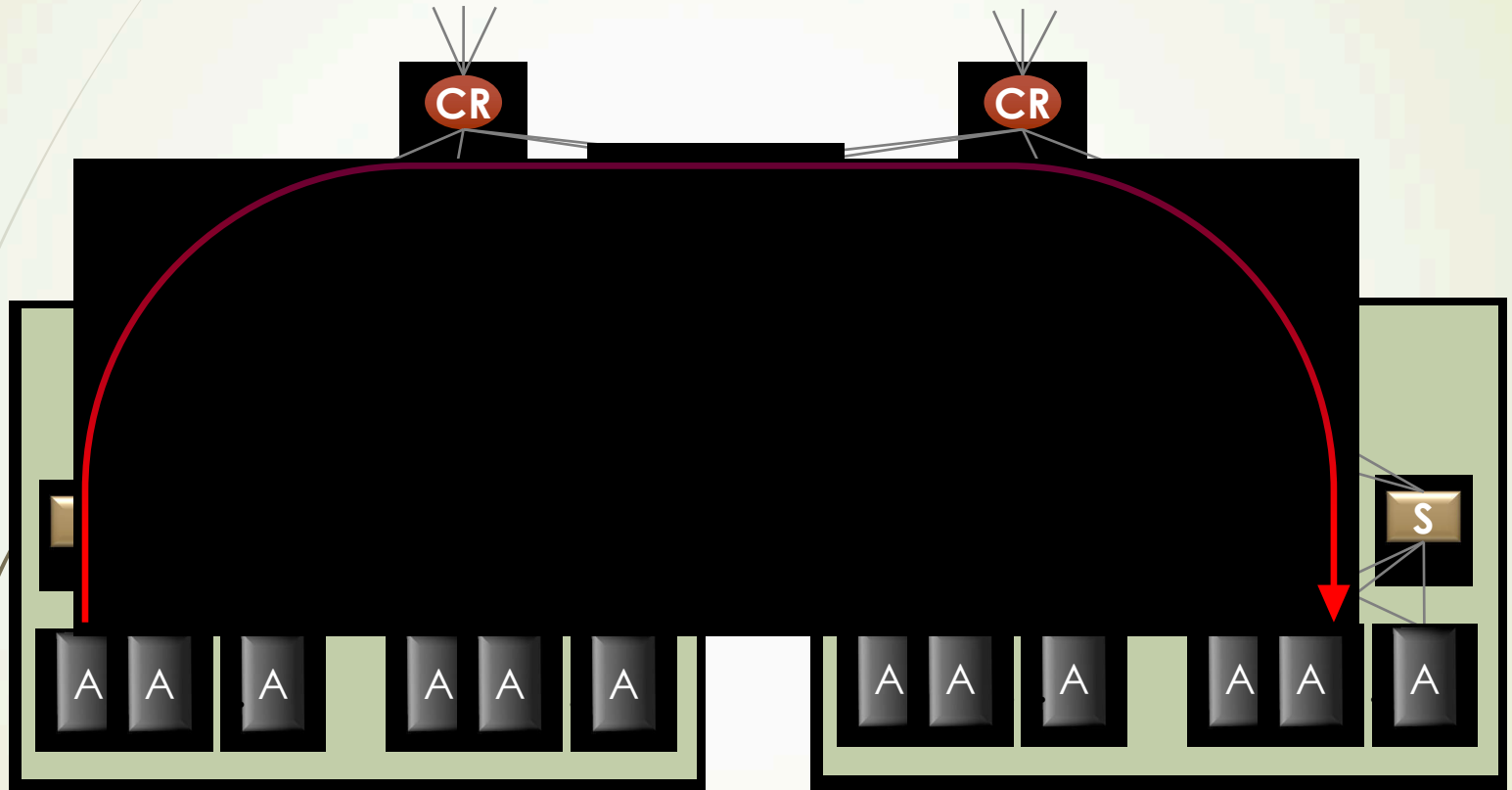
Does It Meet the Requirements?

- Uniform high capacity?
- Performance isolation?
 - typically via VLANs
- Agility in terms of dynamically adding or shrinking servers?
- Agility in terms of adapting to failures, and traffic dynamics?
- Ease of management?

No. There exists a large room for improvement

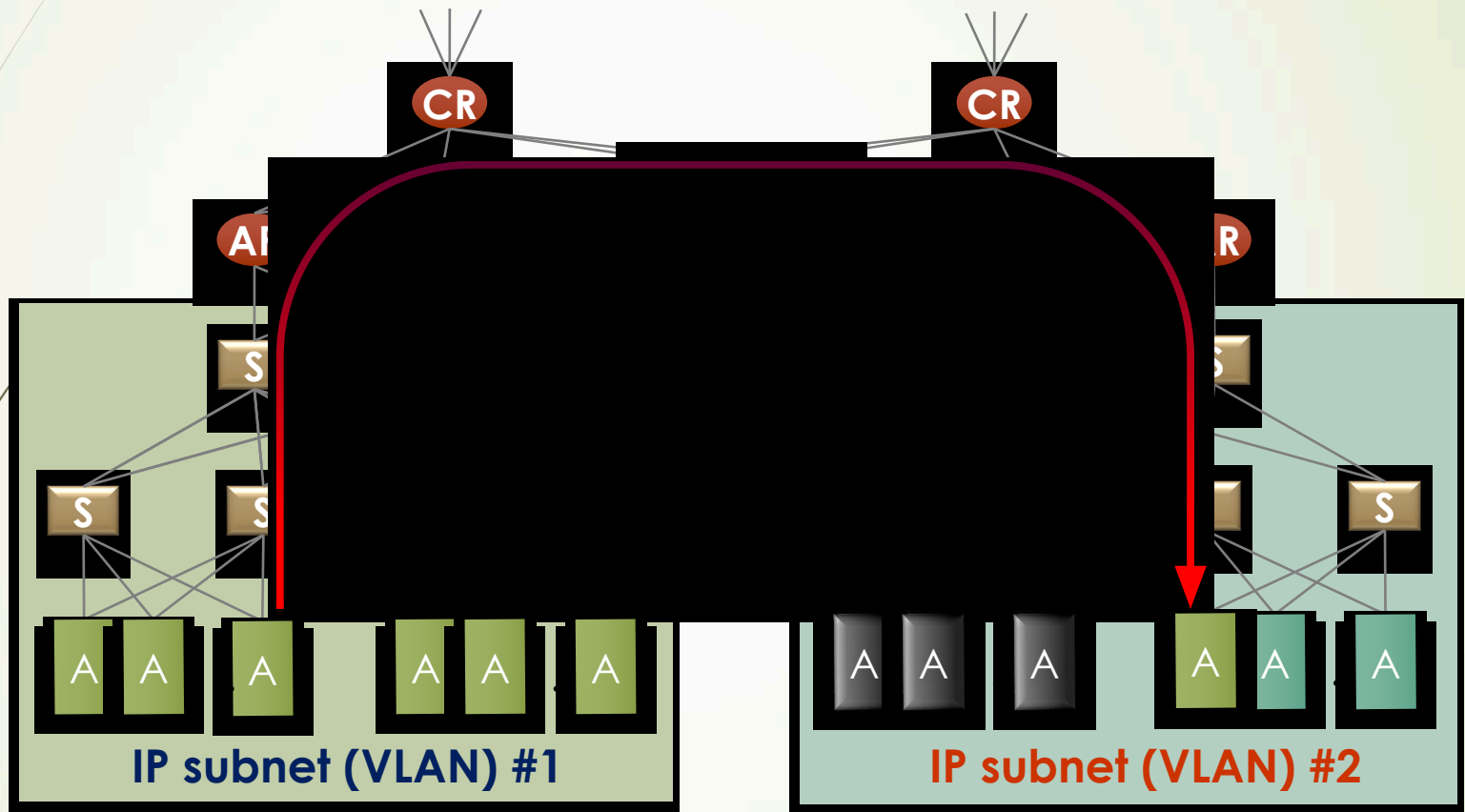


High Oversubscription Ratio



- Extremely limited server-to-server capacity

VLAN-based Isolation



➡ Resource fragmentation

Tedious Configurations

OK, we need something new

